# Welcome

## COURSE:
## ADVANCED PROJECT MANAGEMENT IN AI

## Lesson 4 Session 1

**Risk Management in AI Projects:**
Risk Identification

Welcome

# 课程：
# 人工智能高级项目管理

## 第4课第1节

## AI 项目中的风险管理：
## 风险识别

# LESSON 4.1 – Risk Identification in AI Projects



# 4.1 AI 项目中的风险识别



# Learning Objectives

- Identify major categories of risks unique to AI projects
- Recognize how data, algorithms, and deployment environments introduce uncertainty
- Distinguish technical AI risks from traditional IT project risks
- Apply structured thinking to systematically identify AI project risks early

# 学习目标

- 识别 AI 项目特有的主要风险类别
- 认识数据、算法和部署环境如何引入不确定性
- 区分 AI 技术风险与传统 IT 项目风险
- 运用结构化思维，在早期系统地识别 AI 项目风险

# Risk Identification from a Project Management Perspective

- AI projects involve high uncertainty in data, models, and outcomes
- Early identification of risks prevents costly redesign later
- Many AI failures occur not because the model is incorrect, but because risks were not recognized early
- AI risks extend beyond technology into ethics, regulation, and social impact
- Project managers play a critical role in anticipating cross-disciplinary risks

(PMI, 2021)

# 从项目管理视角看风险识别

- AI 项目在数据、模型和结果方面存在高度不确定性
- 早期识别风险可以避免后期代价高昂的重新设计
- 许多 AI 项目的失败并非因为模型不正确，而是因为风险未能及早被识别
- AI 风险超越了技术范畴，延伸至伦理、法规和社会影响
- 项目经理在预判跨学科风险方面扮演着关键角色

（项目管理协会，2021）

# AI Risk Landscape Compared to Traditional Projects

Traditional software risks:

- Schedule overruns
- Budget constraints
- Integration issues

AI-specific risks:

- Data dependency and data quality issues
- Model unpredictability
- Ethical concerns and bias
- Performance degradation over time

AI projects therefore combine technical, organizational, and societal risks.

(Amershi et al., 2019)

# AI 风险格局与传统项目的比较

传统软件风险：

- 进度超支
- 预算限制
- 集成问题

AI 特有风险：

- 数据依赖性与数据质量问题
- 模型不可预测性
- 伦理问题与偏见
- 性能随时间衰减

因此，AI 项目融合了技术、组织和社会层面的风险。
（阿梅尔希等人，2019）

# Major Risk Categories

AI project risks typically fall into five major areas:

1. Data Risks – poor quality or insufficient data
2. Algorithmic Risks – bias, overfitting, lack of generalization
3. Operational Risks – deployment and system integration issues
4. Lifecycle Risks – model drift and performance decay
5. Regulatory & Ethical Risks – compliance, transparency, accountability

A structured approach to identifying these categories improves AI project planning.

# 主要风险类别

AI 项目风险通常分为五大领域：

1. 数据风险 – 数据质量差或数据不足
2. 算法风险 – 偏见、过拟合、缺乏泛化能力
3. 运营风险 – 部署和系统集成问题
4. 生命周期风险 – 模型漂移和性能衰减
5. 监管与伦理风险 – 合规性、透明度、问责制

采用结构化方法来识别这些类别，可以改进 AI 项目规划。

# Why Data Is the Primary Risk Driver

AI systems learn patterns from data.

Common data risks include:

- Missing or incomplete datasets
- Noisy or inconsistent data
- Incorrect labeling of training data
- Non-representative samples
- Limited historical data

Poor data quality leads to poor model performance regardless of algorithm sophistication.

(Sculley et al., 2015)

# 为何数据是首要风险驱动因素

AI 系统从数据中学习模式。

常见的数据风险包括：

- 数据集缺失或不完整
- 数据存在噪声或不一致
- 训练数据标注错误
- 样本缺乏代表性
- 历史数据有限

无论算法多么复杂，低质量的数据都会导致模型性能不佳。

（斯卡利等人，2015）

# Challenges in Obtaining Usable Data

Even when data exists, access may be restricted.

Common issues include:

- Privacy restrictions (healthcare, finance)
- Data stored across fragmented systems
- Proprietary or third-party data dependencies
- Expensive data acquisition costs
- Delays in data collection pipelines

Data access challenges often become hidden schedule risks in AI projects.

# 获取可用数据的挑战

即使数据存在，访问也可能受到限制。

常见问题包括：

- 隐私限制（医疗、金融领域）
- 数据存储在分散的系统中
- 依赖专有数据或第三方数据
- 高昂的数据获取成本
- 数据收集管道的延迟

在 AI 项目中，数据访问挑战往往会成为隐藏的进度风险。

# Bias in Machine Learning Models

Algorithmic bias occurs when a model produces systematically unfair outcomes for certain groups.

Sources of bias may include:

- Biased historical datasets
- Imbalanced class representation
- Proxy variables reflecting sensitive attributes
- Human labeling biases

Examples include biased hiring algorithms or discriminatory credit scoring systems.

(Amershi et al., 2019)

# 机器学习模型中的偏见

当模型对某些群体产生系统性不公平结果时，就出现了算法偏见。

偏见的来源可能包括：

- 带有偏见的历史数据集
- 类别代表性不均衡
- 反映敏感属性的代理变量
- 人工标注的偏见

例如，带有偏见的招聘算法或存在歧视的信用评分系统。

（阿梅尔希等人，2019）

# Model Performance Risks

AI models may perform well during training but fail in real-world conditions.

Key risks include:

- Overfitting – model memorizes training data rather than learning patterns
- Poor generalization to new data
- Data leakage during training or testing
- Overly complex models with unstable predictions

These risks can cause unexpected failures after deployment.

# 模型性能风险

AI 模型可能在训练时表现良好，但在实际环境中却会失效。

关键风险包括：

- 过拟合 – 模型死记硬背训练数据，而非学习其中的模式
- 对新数据的泛化能力差
- 训练或测试过程中的数据泄露
- 模型过于复杂，导致预测结果不稳定

这些风险可能导致模型在部署后出现意外失效。

# AI Systems Change Over Time

Unlike traditional software, AI systems depend on evolving data.

Types of drift include:

- Data drift – input data distribution changes
- Concept drift – relationship between inputs and outputs changes
- Seasonal or behavioral changes in user patterns

Without monitoring, model performance gradually deteriorates.

(Sculley et al., 2015)

# AI 系统随时间推移而变化

与传统软件不同，AI 系统依赖于不断演化的数据。

漂移的类型包括：

- 数据漂移 – 输入数据分布发生变化
- 概念漂移 – 输入与输出之间的关系发生变化
- 用户模式的季节性变化或行为变化

如果不加以监控，模型性能会逐渐下降。

（斯卡利等人，2015）

# Operational Challenges in AI Deployment

AI models must operate within real systems.

Common deployment risks:

- Integration with legacy systems
- Latency and scalability limitations
- Infrastructure constraints (GPU or cloud capacity)
- Model reproducibility challenges
- Lack of monitoring pipelines

Many AI projects succeed in experimentation but fail during production deployment.

(Amershi et al., 2019)

# AI 部署中的运营挑战

AI 模型必须在实际系统中运行。

常见的部署风险：

- 与遗留系统的集成
- 延迟和可扩展性限制
- 基础设施限制（GPU 或云容量）
- 模型复现性挑战
- 缺乏监控管道

许多 AI 项目在实验阶段取得成功，但在生产部署阶段却失败了。

（阿梅尔希等人，2019）

# Legal and Governance Risks

AI systems increasingly operate within regulatory frameworks.

Potential risks include:

- Violations of privacy laws
- Lack of transparency in automated decisions
- Inability to explain model outputs
- Non-compliance with emerging AI regulations

Industries such as healthcare and finance face particularly high regulatory exposure.

# 法律与治理风险

AI 系统越来越多地在监管框架内运行。

潜在风险包括：

- 违反隐私法
- 自动化决策缺乏透明度
- 无法解释模型输出
- 不符合新兴的 AI 法规

医疗和金融等行业的监管风险尤为突出。

# Key Takeaways:

Project managers may not design algorithms, but they must:

- Facilitate cross-disciplinary risk discussions
- Ensure early identification of technical and ethical risks
- Encourage documentation of assumptions and uncertainties
- Include risk checkpoints throughout the project lifecycle
- Align risk identification with governance frameworks

Effective risk identification enables proactive mitigation rather than reactive problem solving.

# 要点总结：

项目经理或许不设计算法，但他们必须：

- 促进跨学科的风险讨论
- 确保及早识别技术和伦理风险
- 鼓励记录假设和不确定性因素
- 在整个项目生命周期中设置风险检查点
- 使风险识别与治理框架保持一致

有效的风险识别能够实现主动缓解，而非被动解决问题。

# THANKS !!!



# 谢谢！！！



# Welcome

## COURSE:
## ADVANCED PROJECT MANAGEMENT IN AI

## Lesson 4 Session 2

**Risk Management in AI Projects:**
Risk Analysis and Mitigation

Welcome

# 课程：
# 人工智能高级项目管理

## 第4课第2节

## AI 项目中的风险管理：
## 风险分析与缓解

# LESSON 4.2 – Risk Analysis and Mitigation in AI Projects



# 4.2 AI 项目中的风险分析与缓解



# Learning Objectives:

- Differentiate qualitative and quantitative risk analysis approaches in AI projects
- Apply risk matrices to prioritize AI risks
- Understand how data, model, and operational risks influence project outcomes
- Use sensitivity analysis to evaluate AI system uncertainty
- Recognize bias mitigation techniques used in responsible AI development
- Apply structured mitigation strategies within AI project management

# 学习目标

- 区分 AI 项目中的定性与定量风险分析方法
- 运用风险矩阵对 AI 风险进行优先级排序
- 理解数据、模型和运营风险如何影响项目成果
- 使用敏感性分析评估 AI 系统的不确定性
- 识别负责任的 AI 开发中使用的偏见缓解技术
- 在 AI 项目管理中应用结构化的缓解策略

# Why Risk Analysis Matters for Project Managers

AI projects introduce probabilistic uncertainty, meaning outcomes cannot always be predicted with precision.

Unlike traditional IT systems that follow deterministic logic, AI systems learn from data and may behave differently when exposed to new inputs.

For project managers, this means risk management must go beyond identifying risks and move toward structured evaluation and prioritization.

PMs must help teams answer questions such as:

- Which risks are most likely to derail project outcomes?
- Which risks have major ethical or regulatory consequences?
- Where should resources be allocated to mitigation efforts?

Effective AI risk analysis enables PMs to prioritize interventions, allocate budgets, and guide decision making under uncertainty.

# 为什么风险分析对项目经理至关重要

AI 项目引入了概率性不确定性，这意味着其结果并非总能被精确预测。

与遵循确定性逻辑的传统 IT 系统不同，AI 系统从数据中学习，并且在面对新的输入时可能表现出不同的行为。

对项目经理而言，这意味着风险管理必须超越识别风险，转向结构化的评估和优先级排序。

项目经理必须帮助团队解答诸如以下的问题：

- 哪些风险最有可能阻碍项目成果的实现？
- 哪些风险会带来重大的伦理或监管后果？
- 资源应该分配给哪些缓解工作？

有效的 AI 风险分析使项目经理能够在不确定性下确定干预措施的优先级、分配预算并指导决策。

# Risk Analysis in AI Project Management -1

Risk analysis evaluates two core dimensions:

Probability: likelihood that a risk will occur

Impact: severity of consequences if it occurs

AI systems introduce unique evaluation challenges because risks may arise from:

- Data characteristics
- Algorithm behavior
- Deployment environments
- Human decision interaction

# AI 项目管理中的风险分析 - 1

风险分析评估两个核心维度：

概率：风险发生的可能性

影响：风险发生所产生后果的严重程度

AI 系统带来了独特的评估挑战，因为风险可能源于：

- 数据特征
- 算法行为
- 部署环境
- 人类决策的互动

# Risk Analysis in AI Project Management - 2

Therefore, AI project risk analysis typically combines technical evaluation with project management frameworks.

Two primary approaches are commonly used:

- Qualitative Risk Analysis
- Quantitative Risk Analysis

# AI 项目管理中的风险分析 - 2

因此，AI 项目风险分析通常将技术评估与项目管理框架相结合。

常用的两种主要方法是：

- 定性风险分析
- 定量风险分析

# Qualitative Risk Analysis - 1

Qualitative analysis evaluates risks using expert judgment and categorical scales rather than precise numerical models.

Common qualitative tools include:

- Probability–impact matrices
- Risk categorization
- Expert scoring workshops
- Stakeholder risk assessments

# 定性风险分析 - 1

定性分析运用专家判断和分类尺度来评估风险，而非精确的数值模型。

常见的定性分析工具包括：

- 概率-影响矩阵
- 风险分类
- 专家评分研讨会
- 利益相关者风险评估

# Qualitative Risk Analysis - 2

Example in AI projects:

- A team building a medical diagnostic AI system may identify:
- Risk: biased training dataset
- Likelihood: medium
- Impact: high (misdiagnosis)
- The risk is therefore prioritized for mitigation.
- Qualitative analysis is often used early in AI projects when uncertainty is high and detailed data is unavailable.

# 定性风险分析 - 2

AI 项目示例：

- 一个构建医疗诊断 AI 系统的团队可能会识别出：
- 风险：训练数据集存在偏见
- 可能性：中
- 影响：高（可能导致误诊）
- 因此，该风险被优先考虑进行缓解。
- 在 AI 项目早期，当不确定性较高且无法获得详细数据时，通常会使用定性分析。

# Quantitative Risk Analysis - 1

Quantitative analysis evaluates risks using numerical models and statistical techniques.

This approach estimates the expected magnitude of risk impact on project outcomes.

Common techniques include:

- Monte Carlo simulations
- Sensitivity analysis
- Expected monetary value (EMV)
- Statistical performance analysis

# 定量风险分析 - 1

定量分析运用数值模型和统计技术来评估风险。

这种方法旨在估算风险对项目成果可能产生的影响程度。

常用技术包括：

- 蒙特卡洛模拟
- 敏感性分析
- 预期货币价值
- 统计性能分析

# Quantitative Risk Analysis - 2

In AI projects, quantitative analysis may assess:

- Financial losses from incorrect predictions
- Operational disruption due to model errors
- Probability of performance falling below thresholds

These methods provide data-driven insights to support strategic decision making.

# 定量风险分析 - 2

在 AI 项目中，定量分析可用于评估：

- 因错误预测导致的财务损失
- 模型错误造成的运营中断
- 性能低于阈值的概率

这些方法提供数据驱动的洞察，以支持战略决策。

# Risk Matrix for AI Projects - 1

A risk matrix helps project managers prioritize risks by combining probability and impact scores.




Typical matrix categories include:

- Low Risk – minimal monitoring required
- Moderate Risk – mitigation planning needed
- High Risk – immediate management attention required

# AI 项目风险矩阵 - 1

风险矩阵通过结合概率和影响评分，帮助项目经理对风险进行优先级排序。




典型的矩阵类别包括：

- 低风险 – 只需最低限度的监控
- 中等风险 – 需要进行缓解规划
- 高风险 – 需要管理层立即关注

# Risk Matrix for AI Projects - 2

Example AI risks placed in a matrix:

High impact / high probability

- Data bias affecting model predictions

High impact / low probability

- Regulatory non-compliance

Low impact / high probability

- Minor data pipeline delays

The matrix allows teams to focus mitigation resources on the most critical risks first.

# AI 项目风险矩阵 - 2

以下是放置在矩阵中的 AI 风险示例：

高影响 / 高概率

- 影响模型预测的数据偏见

高影响 / 低概率

- 监管违规

低影响 / 高概率

- 次要数据管道延迟

该矩阵使团队能够将缓解资源优先集中于最关键的风险上。

# Sensitivity Analysis in AI Systems

Sensitivity analysis examines how changes in input variables affect model outputs.

This is particularly important in AI systems where small variations in data may significantly change predictions.

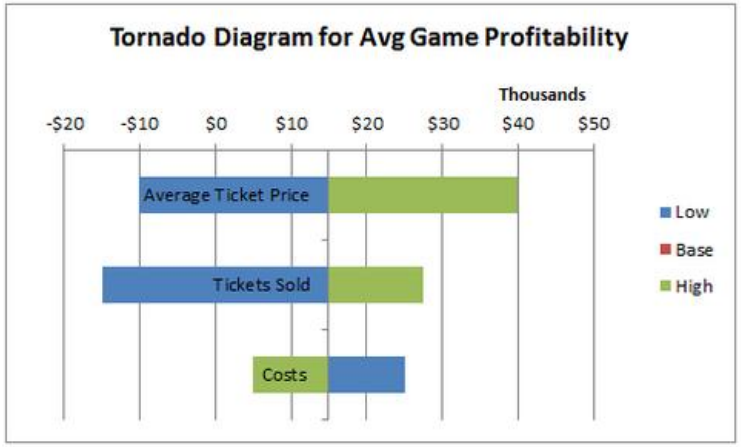
Project managers use sensitivity analysis to identify:

- Variables that strongly influence model outcomes
- Data features that create instability
- Areas where additional validation is required

For example, a credit risk AI model may show strong sensitivity to income data.

If that variable contains inaccuracies, predictions may become unreliable.

Sensitivity analysis therefore helps prioritize data quality improvement efforts.




https://en.wikipedia.org/wiki/Tornado_diagram

# AI 系统中的敏感性分析

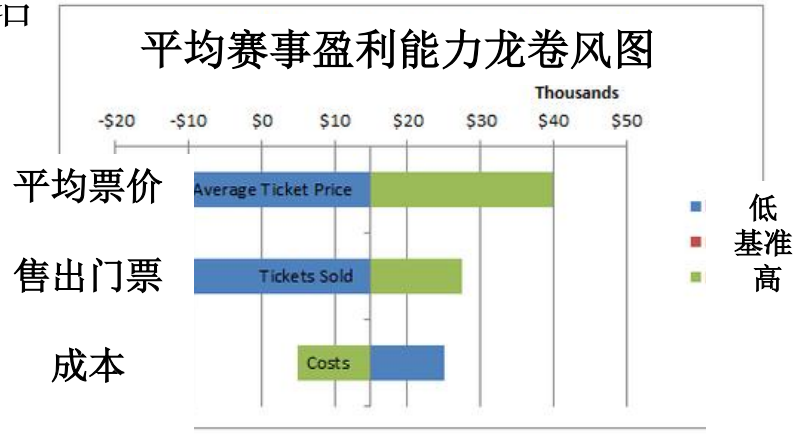
敏感性分析用于检验输入变量的变化如何影响模型输出。

这在 AI 系统中尤其重要，因为数据的微小变化都可能导致预测结果的显著改变。

项目经理利用敏感性分析来识别：

- 对模型结果影响强烈的变量
- 导致不稳定的数据特征
- 需要额外验证的领域

例如，一个信用风险 AI 模型可能对收入数据表现出强敏感性。
如果该变量包含不准确信息，预测结果就可能变得不可靠。
因此，敏感性分析有助于优先安排数据质量改进工作。




https://en.wikipedia.org/wiki/Tornado_diagram

# Bias Risks in AI Systems -1

Algorithmic bias occurs when AI systems produce systematically unfair outcomes for certain groups.

Bias can originate from:

- Historical data reflecting societal inequalities
- Unbalanced training datasets
- Feature selection choices
- Algorithm design decisions

# AI 系统中的偏见风险 - 1

当 AI 系统对某些群体产生系统性不公平结果时，即出现算法偏见。

偏见可能源于：

- 反映社会不平等的历史数据
- 不平衡的训练数据集
- 特征选择
- 算法设计决策

# Bias Risks in AI Systems - 2

For project managers, bias risks are not only technical issues but also ethical, legal, and reputational risks.

Examples of real-world consequences include:

- discriminatory hiring algorithms
- biased credit scoring systems
- inaccurate medical diagnosis for certain populations

Managing bias is therefore a core risk mitigation responsibility in AI projects.

(O'Neil, 2016)

# AI 系统中的偏见风险 - 2

对项目经理而言，偏见风险不仅是技术问题，更是伦理、法律和声誉风险。

现实世界中的后果实例包括：

- 歧视性的招聘算法
- 存在偏见的信用评分系统
- 针对某些人群的不准确的医疗诊断

因此，管理偏见是 AI 项目中风险缓解的核心职责之一。

（奥尼尔，2016）

# Bias Mitigation Strategies - 1

Bias mitigation can be implemented at multiple stages of the AI lifecycle.

1. Pre-processing methods

Improving training data before model development

Examples:

- balancing datasets
- removing sensitive attributes
- synthetic data generation

# 偏见缓解策略 - 1

偏见缓解可以在 AI 生命周期的多个阶段实施。

1. 预处理方法

在模型开发前改进训练数据

示例：

- 平衡数据集
- 移除敏感属性
- 生成合成数据

# Bias Mitigation Strategies - 2

2. In-processing methods

Adjusting algorithms during training

Examples:

- fairness-aware machine learning models
- constrained optimization

3. Post-processing methods

Correcting outputs after model predictions

Examples:

- recalibrating predictions
- fairness adjustments across groups

These strategies help ensure AI systems produce equitable outcomes.

# 偏见缓解策略 - 2

2. 处理中方法

在训练过程中调整算法

示例：

- 公平性感知的机器学习模型
- 约束优化

3. 后处理方法

在模型预测后修正输出

示例：

- 重新校准预测结果
- 跨群体公平性调整

这些策略有助于确保 AI 系统产生公平的结果。

# Ethical AI Frameworks for Risk Mitigation - 1

Organizations increasingly adopt ethical AI frameworks to guide risk mitigation.

Key principles commonly emphasized include:

- fairness
- transparency
- accountability
- safety
- privacy protection

# 用于风险缓解的伦理 AI 框架 - 1

各组织越来越多地采用伦理 AI 框架来指导风险缓解工作。

通常强调的关键原则包括：

- 公平性
- 透明度
- 问责制
- 安全性
- 隐私保护

# Ethical AI Frameworks for Risk Mitigation - 2

Examples of widely referenced frameworks include:

- OECD AI Principles
- Responsible AI guidelines from major technology firms
- national AI governance strategies

These frameworks provide structured guidance for managing ethical risks throughout the AI lifecycle.

(OECD, 2019)

# 用于风险缓解的伦理 AI 框架 - 2

被广泛参考的框架示例包括：

- OECD AI 原则
- 主要科技公司的负责任 AI 指南
- 国家 AI 治理战略

这些框架为在整个 AI 生命周期中管理伦理风险提供了结构化指导。

（经合组织，2019）

# Mitigation Strategies for AI Project Risks - 1

Risk mitigation strategies can be categorized into several approaches:

Avoidance

- Changing project scope or approach to eliminate risk.
- Example: using simpler models instead of high-risk black-box systems.

Reduction

- Implementing controls to lower risk probability.
- Example: improving data validation pipelines.

# AI 项目风险的缓解策略 - 1

风险缓解策略可分为几种方法：

规避

- 改变项目范围或方法来消除风险。
- 示例：使用更简单的模型，而非高风险的黑箱系统。

降低

- 实施控制措施以降低风险概率。
- 示例：改进数据验证管道。

# Mitigation Strategies for AI Project Risks - 2

Transfer

- Sharing risk through partnerships or external services.
- Example: using managed cloud AI services.

Acceptance

- Acknowledging risk when mitigation costs exceed benefits.
- Project managers must select the most appropriate mitigation strategy based on project constraints.

# AI 项目风险的缓解策略 - 2

转移

- 通过合作或外部服务分担风险。
- 示例：使用托管云 AI 服务。

接受

- 当缓解成本超过收益时，承认风险的存在。
- 项目经理必须根据项目约束条件选择最合适的缓解策略。

# Integrating Risk Mitigation into AI Project Management

Effective risk mitigation requires integration across the entire project lifecycle.

Key PM practices include:

- continuous risk monitoring
- periodic model performance reviews
- stakeholder risk communication
- governance checkpoints before deployment

Risk mitigation should also be embedded within Agile AI development cycles, where risks are reassessed after each iteration.

This ensures that AI systems remain reliable, ethical, and aligned with business objectives.

# 将风险缓解融入 AI 项目管理

有效的风险缓解需要贯穿整个项目生命周期进行整合。

项目经理的关键实践包括：

- 持续的风险监控
- 定期的模型性能审查
- 与利益相关者的风险沟通
- 部署前的治理检查点

风险缓解还应嵌入敏捷 AI 开发周期中，在每次迭代后重新评估风险。这能确保 AI 系统保持可靠、合乎伦理，并与业务目标保持一致。

# Key Takeaways

- AI projects require structured risk analysis beyond simple risk identification.
- Qualitative and quantitative methods help prioritize risks effectively.
- Risk matrices enable clear prioritization for project decision making.
- Sensitivity analysis reveals how input variables influence AI system behavior.
- Algorithmic bias represents a critical ethical and operational risk.
- Successful AI projects combine technical mitigation strategies with governance frameworks.

# 要点总结

- AI 项目需要超越简单风险识别的结构化风险分析。
- 定性与定量方法有助于有效确定风险的优先级。
- 风险矩阵能够为项目决策提供清晰的优先级排序。
- 敏感性分析揭示了输入变量如何影响 AI 系统行为。
- 算法偏见是关键的伦理与运营风险。
- 成功的 AI 项目将技术缓解策略与治理框架相结合。

# THANKS !!!



# 谢谢！！！



# Welcome

## COURSE:
## ADVANCED PROJECT MANAGEMENT IN AI

## Lesson 4 Session 3

**Risk Management in AI Projects:**
Governance and Compliance

Welcome

课程

# 人工智能高级项目管理

## 第4课第3节

## AI 项目中的风险管理：治理与合规

# LESSON 4.3 – Governance and Compliance in AI Projects



# 4.3 AI 项目中的治理与合规



# Learning Objectives

- Understand the importance of AI governance and compliance.
- Explore ethical principles guiding AI development.
- Examine explainability (XAI) and accountability frameworks.
- Identify key regulations affecting AI projects.
- Apply responsible AI frameworks in project management.

# 学习目标

- 理解 AI 治理与合规的重要性。
- 探索指导 AI 发展的伦理原则。
- 审视可解释性与问责制框架。
- 识别影响 AI 项目的关键法规。
- 在项目管理中应用负责任的 AI 框架。

# Why Governance and Compliance Matter (PM Perspective)

- Governance ensures AI projects are aligned with organizational values and societal norms, reducing legal, ethical, and reputational risks.

- For PMs, understanding compliance is essential to building stakeholder trust, supporting decision-making under uncertainty, and ensuring sustainable deployment of AI solutions.

# 为何治理与合规至关重要（项目经理视角）

- 治理确保 AI 项目与组织价值观和社会规范保持一致，从而降低法律、伦理和声誉风险。
- 对项目经理而言，理解合规性对于建立利益相关者信任、支持不确定性下的决策以及确保 AI 解决方案的可持续部署至关重要。

# Core AI Ethics Principles

- Transparency: AI decisions must be understandable.
- Fairness: Avoid bias and ensure equitable outcomes.
- Accountability: Clear responsibility for AI outcomes.
- Privacy: Safeguard sensitive data.

OECD AI Principles (OECD, 2019)

# 核心 AI 伦理原则

- 透明度：AI 的决策必须是可理解的。
- 公平性：避免偏见，确保结果公平。
- 问责制：明确 AI 结果的责任归属。
- 隐私：保护敏感数据。

经合组织 AI 原则（经合组织，2019）

# Global Regulatory Landscape

- EU AI Act: Introduces a risk-based framework for AI systems.
- GDPR: Ensures data privacy and protection.
- Sector-specific regulations (healthcare, finance).
- Early compliance planning reduces costly rework and accelerates approvals.

# 全球监管格局

- 欧盟《AI 法案》：引入了基于风险的 AI 系统监管框架。
- 通用数据保护条例：确保数据隐私与保护。
- 特定行业法规（医疗、金融）。
- 尽早进行合规规划可减少代价高昂的返工，并加快审批进程。

# Explainable AI (XAI) in Practice

- Explainable AI techniques help stakeholders understand model outputs, aiding debugging, trust, and regulatory compliance.
- Common tools: SHAP, LIME, attention visualizations.
- PMs use XAI to communicate results effectively to executives and non-technical teams.

# 可解释性 AI 的实践

- 可解释的 AI 技术有助于利益相关者理解模型输出，从而辅助调试、建立信任并确保监管合规。
- 常用工具：SHAP、LIME、注意力可视化。
- 项目经理利用可解释性 AI 向高管和非技术团队有效沟通结果。

# Accountability Structures

- Assigning roles and responsibilities for AI outcomes.
- Establishing internal governance committees for oversight.
- Maintaining documentation of decision-making.
- Audit trails for model development and deployment.

# 问责制结构

- 为 AI 结果分配角色与职责
- 建立内部治理委员会进行监督
- 维护决策过程的文档
- 为模型开发与部署建立审计追踪

# Responsible AI Frameworks

- Frameworks from leading tech organizations focus on fairness, transparency, safety, privacy, and robustness.

- PMs integrate these into risk registers, project checkpoints, and review cycles to ensure ethical deployment.

# 负责任的 AI 框架

- 领先科技公司的框架侧重于公平性、透明度、安全性、隐私性和稳健性。
- 项目经理需将这些框架整合到风险登记册、项目检查点和评审周期中，以确保合乎伦理的部署。

# Detecting and Mitigating Bias

- Identify algorithmic bias early in model development.
- Mitigation techniques: data balancing, fairness-aware modeling.
- Conduct regular audits and validate against diverse datasets.
- PM role: Ensure bias mitigation steps are planned, monitored, and reported.

# 检测与缓解偏见

- 在模型开发早期识别算法偏见。
- 缓解技术：数据平衡、公平性感知建模。
- 定期进行审计，并使用多样化数据集进行验证。
- 项目经理角色：确保偏见缓解步骤得到规划、监控和报告。

# Compliance Across Project Phases

- Data acquisition: Verify consent and privacy compliance.
- Model development: Document assumptions and limitations.
- Deployment: Monitor adherence to ethical and legal standards.
- Integrating compliance checks in each phase reduces risks and fosters trust.

# 项目各阶段的合规性

- 数据获取：核实同意与隐私合规性。
- 模型开发：记录假设与局限性。
- 部署：监控是否符合伦理与法律标准。
- 在每个阶段整合合规检查，可降低风险并增进信任。

# Supporting Tools and Platforms

- Automated compliance tools: data lineage, audit tracking.
- Model monitoring platforms: drift detection, fairness dashboards.
- Collaboration platforms for transparency.
- Choosing the right tools ensures efficiency, accountability, and smoother audits.

# 支持工具与平台

- 自动化合规工具：数据血缘、审计追踪。
- 模型监控平台：漂移检测、公平性仪表板。
- 提升透明度的协作平台。
- 选择合适的工具可确保效率、问责制，并使审计更顺畅。

# Challenges in Governance

- Rapidly evolving regulations.
- Complexity of AI systems.
- Coordinating interdisciplinary teams.
- Best practices: Embed ethics early, maintain documentation, establish cross-functional oversight, and provide continuous training for AI teams.

# 治理中的挑战

- 法规的快速演变
- AI 系统的复杂性
- 跨学科团队的协调
- 最佳实践：尽早融入伦理考量，维护文档，建立跨职能监督机制，并为 AI 团队提供持续培训。

# Key Takeaways

- Governance and compliance are essential for AI project success.
- Ethical principles guide responsible AI development.
- Explainability and accountability build stakeholder trust.
- Regulatory awareness prevents costly risks.
- PMs play a central role in embedding responsible practices.

# 要点总结

- 治理与合规对 AI 项目成功至关重要。
- 伦理原则指导负责任的 AI 发展。
- 可解释性与问责制建立利益相关者的信任。
- 对法规的认知可避免代价高昂的风险。
- 项目经理在融入负责任实践方面发挥着核心作用。

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# THANKS !!!



# 谢谢！！！

